Experiment 2 – Optoelectronic energy conversion (solar cell)

Preparation: tasks, evaluation plan and worked examples

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(Antestat 14:00, building LT, room LT 227)

Quoted from the lab manual *Optoelectronic energy conversion* (e_poe_exp2.pdf). This manual has no numbered questions; it has the experiment tasks (a)–(e) (§5), seven evaluation items I–VII (§6) and the theory (§2) that you must be able to derive. Quotes are in shaded boxes with a teal bar, the worked answers and formulas follow. Every section ends with a short Turkish summary (**TR**). An Excel evaluation template (*06_Auswertung*) calculates all parameters from your measured I-U tables.

0. Background in one page

- **Photovoltaics:** absorbed photons ($E \geq W_G$) create electron-hole pairs; the field of the pn junction separates them → photocurrent $I_{SC} \propto$ irradiance; an external load turns it into electrical power.
- **Radiometry:** irradiance E in W/m²; spectral irradiance in W m^{-2} nm^{-1} (1 µW/(cm²·nm) = 0.01 W/(m²·nm)); AM1.5G: 1000 W/m², standard test conditions (STC): AM1.5G, 1000 W/m², 25 °C.
- **Dark curve (diode):** $I_D = I_S(e^{U/(nU_T)} - 1)$, $U_T = kT/e \approx 26$ mV; $I_S(T) \propto \exp(-W_G/(n\,k\,T))$ – rises very fast with temperature.
- **Light curve:** $I = I_{SC} - I_D$ (eq. 7); with R_S, R_P: $I = I_{SC} - I_S(e^{(U+IR_S)/(nU_T)} - 1) - (U+IR_S)/R_P$.
- **Characteristic numbers:** $I_{SC} = \eta_{ext}\,(e/h\nu)\,P_{opt}$ (eq. 8); $U_{OC} = nU_T \ln(I_{SC}/I_S + 1)$ (eq. 10); MPP: $P_{MPP} = U_{MPP}\,I_{MPP}$; **$FF = P_{MPP}/(U_{OC}\,I_{SC})$**; **$\eta = P_{MPP}/P_{opt} = FF \cdot U_{OC} \cdot I_{SC}/P_{opt}$**.
- **Losses:** reflection, non-absorption ($E < W_G$), thermalisation ($E > W_G$), recombination, series/parallel resistance.
- **Instruments:** LED solar simulator (960 LEDs, warm + cold white, ≤ 300 W/m², PWM-dimmed), spectrometer getSpec-3648 (360–1100 nm, 2.4 nm), Keysight/Keithley B2901A source-measure unit, a fan with two solar cells + LED bypass diodes.

Task (a) – Spectral irradiance of the sun simulator

> **Task (a) (§5):** Use the spectrometer to measure the spectral irradiance of the solar simulator in µW/(cm²·nm).

Evaluation items I and II use this spectrum: (I) compare it with AM1.5G; (II) integrate it over wavelength and compare with the measured irradiance and with 1000 W/m²:

$$E = \int E_\lambda(\lambda)\,d\lambda \approx \Sigma\, E_{\lambda,i} \cdot \Delta\lambda_i \qquad [\mu W/cm^2] \cdot 0.01 = [W/m^2]$$

Unit conversion	Value
1 µW/cm²	0.01 W/m²
1 mW/cm²	10 W/m²
100 mW/cm² (= 1 sun)	1000 W/m²
simulator maximum	300 W/m² = 30 mW/cm² = 30 000 µW/cm²

Example: an average of 75 µW/(cm²·nm) over 400 nm width (400–800 nm) gives 30 000 µW/cm² = 300 W/m². **Expected differences to the sun:** the white LED spectrum has a blue peak (≈ 450 nm) from the LED chip and a broad phosphor band (500–700 nm) but **almost nothing above ≈ 780 nm**, while AM1.5G continues smoothly into the infrared. A silicon cell (band edge 1107 nm) therefore uses less of the simulator spectrum than of sunlight (spectral mismatch) – for a 5900 K black body ≈ 80 % of the power lies below the Si band edge (Fig. A).

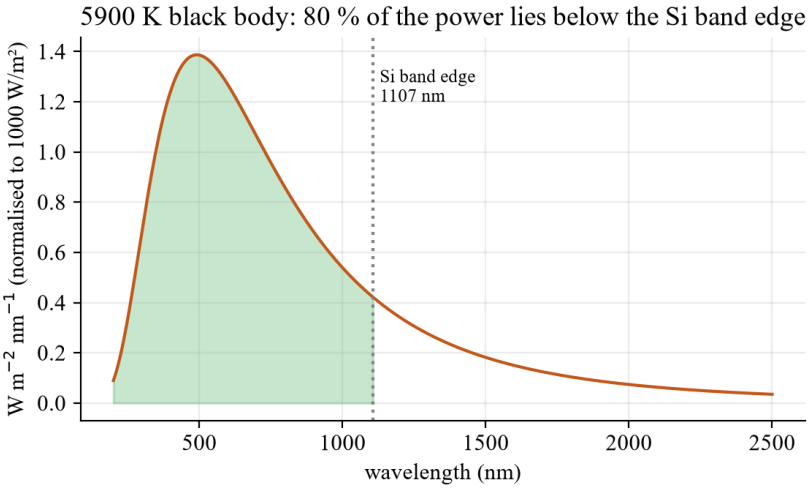


Fig. A – Black-body spectrum 5900 K (≈ solar spectrum without absorption), normalised to 1000 W/m²; green: part below the Si band edge.

TR E = Σ E_λ·Δλ; 1 µW/cm² = 0,01 W/m². LED simülatörü maviye ve 500–700 nm'ye sahip, 780 nm üstünde neredeyse hiç yok → Si pili güneşe göre daha az kullanır (spektral uyumsuzluk).

Task (b) – Area of the solar cell

Task (b) (§5): Determine the area of the solar cell to be measured.

Measure the *active* area A (length × width with a ruler/caliper); decide and **state** whether contact fingers/busbars are included. A is needed for the incident power $P_{opt} = E \cdot A$ and for the current density $J_{SC} = I_{SC}/A$ (A/cm²), which allows comparing cells of different size. Example: 2 cm × 2 cm = 4 cm² → at 100 mW/cm²: P_{opt} = 400 mW; at 30 mW/cm² (max. of the simulator): 120 mW.

TR Aktif alanı ölç (parmak/busbar dahil mi, yaz): P_opt = E·A, J_SC = I_SC/A.

Task (c) – Dark I-U curves at different temperatures

Task (c) (§5): Record I-U characteristics (dark) at different temperatures. Note: Experiments (c) and (d) must be planned (with the help of the supervisor) in such a way that reliable statements can be made about the band gap of the semiconductor material used and the intensity dependence of the power efficiency.

Plan: at least 4–5 temperatures (e.g. 20, 30, 40, 50, 60 °C), wait for thermal equilibrium at each, sweep U from ≈ 0 to ≈ 0.7 V (forward) with fine steps (the current spans 6–8 decades – use a log-friendly step), plus a short reverse sweep (−0.5 … 0 V) for R_P. Keep the cell **dark** and the compliance current below the cell rating.

Evaluation item III (dark parameters):

1. **Semi-log plot** ln I (or log I) versus U: in the exponential range a straight line: slope = $1/(n\,U_T)$ → **$n = 1/(\text{slope} \cdot U_T)$**, intercept → **$I_S = e^{\text{intercept}}$** ($U_T$ = kT/e from the measured cell temperature in K).
2. **Series resistance R_S:** at high current the curve bends away from the straight line; $R_S \approx \Delta U_{deviation}/I$ (or the slope dU/dI in the linear region at the highest currents).
3. **Parallel resistance R_P:** at small |U| (and in reverse bias) the curve is ohmic: $R_P = \Delta U/\Delta I$ near U = 0 (the shunt leakage hides the diode current there).
4. **Band gap W_G (eq. 6):** $I_S(T) \propto T^3 \exp(-W_G/(nkT))$; plot **ln I_S versus 1/T** (T in K) for all temperatures: slope = $-W_G/(n\,k)$ → **$W_G = -n \cdot k \cdot \text{slope}$** ($k = 8.617 \cdot 10^{-5}$ eV/K); correct for the T^3 prefactor by plotting $\ln(I_S/T^3)$ if you want a better fit. Si: W_G = 1.12 eV.

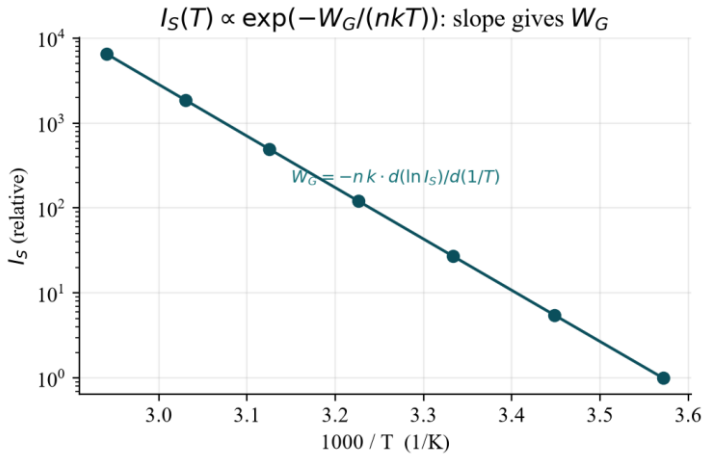


Fig. B – $I_S(T)$ on a logarithmic axis versus 1000/T (model for W_G = 1.12 eV): a straight line, the slope gives the band gap.

Why does the dark curve move to lower voltage when T rises? I_S grows exponentially with T ($\exp(-W_G/nkT)$), so the same current is reached at a lower U (≈ −2 mV/K) – this is also why U_{OC} falls with temperature. **Errors:** the measured temperature is the cell stage, not the junction; self-heating at high currents; leakage at small U; n and W_G are strongly coupled (use the n from the same fit).

TR ln I – U doğrusunun eğimi 1/(n·U_T) → n; kesişim → I_S. R_S yüksek akımda sapma, R_P U≈0'da omik eğim. ln I_S – 1/T doğrusunun eğimi −W_G/(n·k) → W_G (Si: 1,12 eV). T ↑ → I_S ↑ → eğri sola kayar, U_OC düşer.

Task (d) – Bright I-U curves at different intensities

Task (d) (§5): Record I-U characteristics (bright) at different intensities.

Intensities 25 %, 50 %, 75 % and 100 % of the simulator (PWM duty cycle); constant cell temperature (use short sweeps; at 300 W/m² the cell heats – let it cool between runs). Sweep from U ≤ 0 (reverse, shows I_{SC} region) to beyond U_{OC} (≈ 0.6 V for Si, a bit more for 100 %). **P_{opt} for every intensity:** P_{opt} = E·A with the irradiance E from task (a) (scaled with the duty cycle) and the area A from task (b).

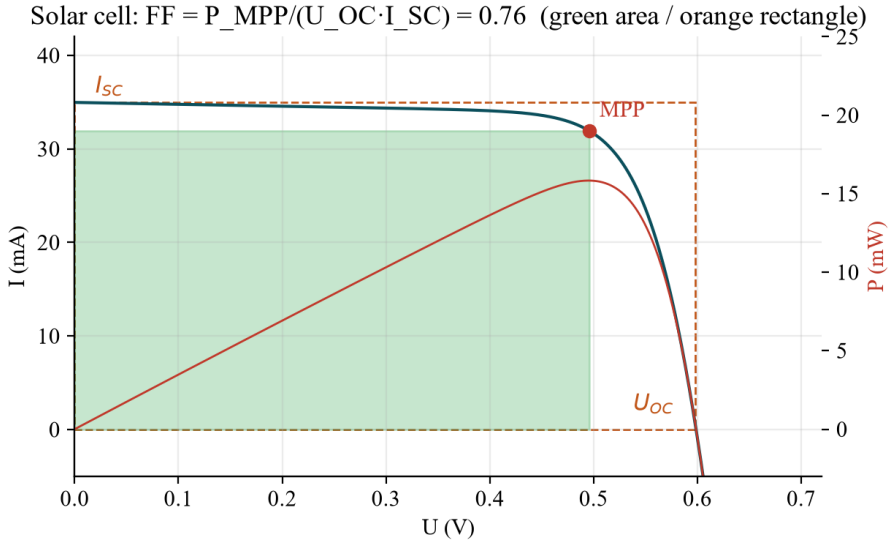


Fig. C – Bright I-U curve with I_{SC}, U_{OC}, the maximum power point and the fill factor (model: Si, 1 cm², I_{SC} = 35 mA, R_S = 0.5 Ω, R_P = 500 Ω).

Evaluation item IV: read from each curve **I_{SC}** (current at U = 0), **U_{OC}** (voltage at I = 0, interpolate the zero crossing), **P = U·I** for every point → **P_{MPP}** (maximum) with **U_{MPP}, I_{MPP}, P_{opt} = E·A**,

$$FF = P_{MPP} / (U_{OC} \cdot I_{SC}) \qquad \eta = P_{MPP} / P_{opt} = FF \cdot U_{OC} \cdot I_{SC} / P_{opt}$$

Worked example (the model of Fig. C, 1 cm², 100 mW/cm²): I_{SC} = 35.0 mA, U_{OC} = 0.598 V, P_{MPP} = 15.8 mW at U_{MPP} = 0.50 V → FF = 15.8/(0.598·35.0) = **0.76**, η = 15.8 mW/100 mW = **15.8 %**.

Evaluation item V – intensity dependence (eq. 13–14)

At C-fold intensity $P_{opt} \rightarrow C \cdot P_{opt}$, the generated carriers and $I_{SC} \rightarrow C \cdot I_{SC}$ (linear). From eq. (10) $U_{OC} \rightarrow n\, U_T \ln(C \cdot I_{SC}/I_S + 1) \approx U_{OC} + n\, U_T \ln C$ (about +70 mV per decade for n = 1.2). With FF roughly constant,

$$\eta(C)/\eta(1) = U_{OC}(C)/U_{OC}(1) \approx 1 + n\, U_T \ln C \,/\, U_{OC}(1)$$

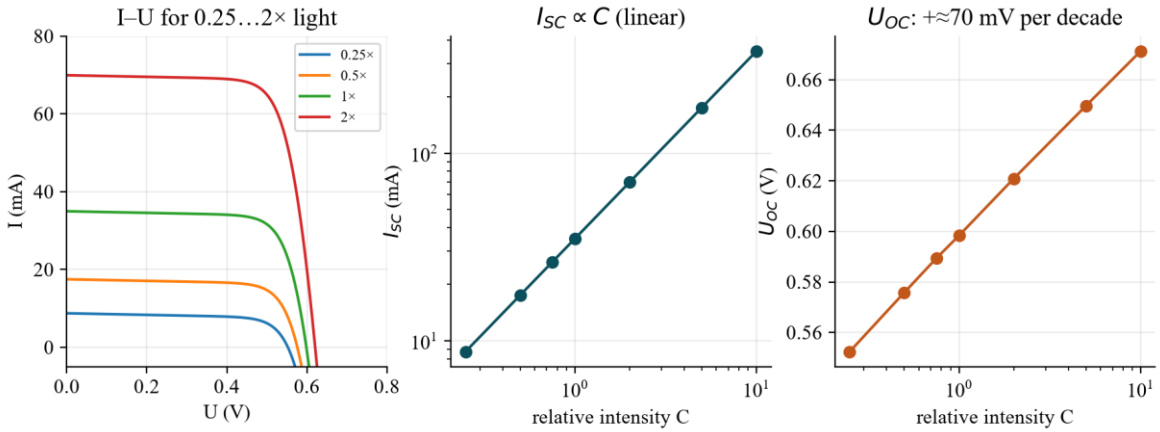


Fig. D – Intensity dependence (model): I_{SC} is proportional to C, U_{OC} grows only logarithmically; hence the efficiency rises slowly with intensity.

C (relative intensity)	I_SC (mA)	U_OC (V)	P_MPP (mW)	FF	η (%)
0.25	8.7	0.552	3.4	0.70	13.6
0.50	17.5	0.576	7.5	0.74	15.0
0.75	26.2	0.589	11.6	0.75	15.5
1.00	35.0	0.598	15.8	0.76	15.8
2.00	69.9	0.621	32.5	0.75	16.2

Discussion for V: η rises with intensity (logarithmically, the U_{OC} term), flattens and finally **falls at high intensity** because the series resistance loss I^2R_S grows with C^2 (FF drops, Fig. E left) and the cell heats. At low intensity the shunt resistance R_P (leakage) and recombination reduce U_{OC} and FF (right) – so the efficiency is smaller in the weak-light winter months. **The LTspice demo reproduces these curves:** (see below).

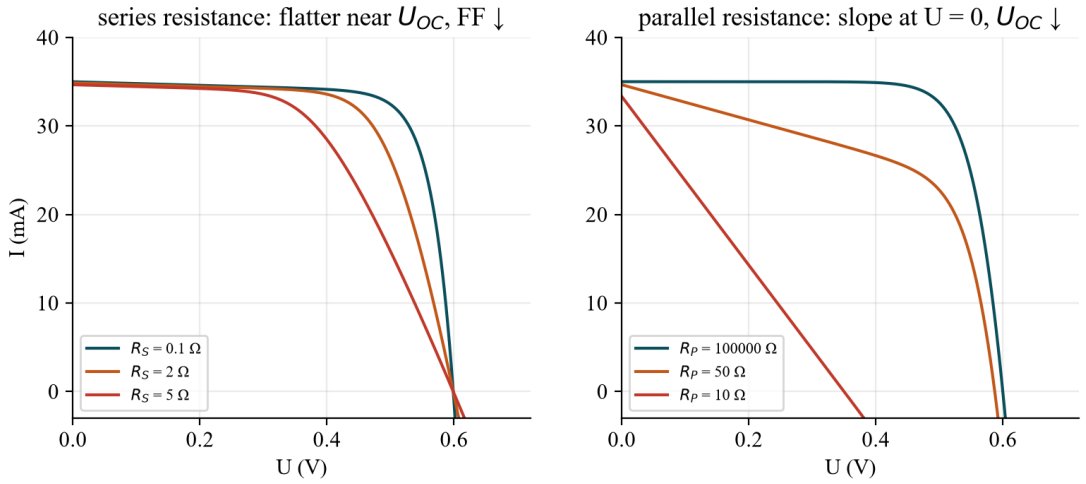


Fig. E – Effect of the series and the parallel resistance on the light curve.

TR η = FF·U_OC·I_SC/P_opt; I_SC ∝ C, U_OC ≈ U_OC(1) + n·U_T·ln C → η yavaşça artar, yüksek şiddette R_S (I²R_S) ve ısınma yüzünden düşer; düşük ışıkta R_P/rekombinasyon U_OC ve FF'yi düşürür.

Task (e) – Solar cells with fan and bypass LEDs

Task (e) (§4/§5): Set up the circuits described in section 4 and investigate the effects of a change in intensity (25 %, 50 %, 75 % and 100 %) on the speed of rotation of the fan and the brightness of the LEDs (bypass diodes). (§4: a solar cell and a light-emitting diode are connected in parallel, whereby the polarity of the components must be observed; the fan is connected in series with this circuit; a second parallel circuit of solar cell and LED is added – Fig. 3-9.)

- **Circuit:** [solar cell 1 ∥ LED 1] in series with [solar cell 2 ∥ LED 2] in series with the fan. Each LED is connected **antiparallel** to its cell (polarity!): it conducts only if the cell voltage reverses.
- **Equal illumination:** both cells deliver current to the fan ($\approx I_{SC} \propto$ intensity); the LEDs stay dark, the fan speed rises with the intensity (it is driven by the voltage/current available: the fan stands still at low intensity until the current overcomes friction).
- **One cell shaded/weaker:** in the series connection the weakest cell limits the current. The stronger cell wants to push a larger current than the shaded cell can give – the surplus flows through the **bypass diode (LED)** of the shaded cell: the LED lights up (brightness ∝ current), the shaded cell is protected from being forced into reverse breakdown/hot spots, and the fan keeps running with a reduced voltage.
- **Record:** a table with 25/50/75/100 % × (both cells lit / one cell shaded): fan speed (qualitative: stops / slow / medium / fast or counted rpm), LED 1 and 2 brightness, and the measured voltage/current if possible.

Evaluation item VI: summarise the observations in tabular form and discuss the visual impression at the different illuminance levels. **Why a bypass diode?** Without it a shaded cell in a series string becomes a high-resistance load and is driven in reverse (hot-spot heating, lost power of the whole string); with it the string loses only the voltage of the shaded cell.

TR LED, pile antiparalel bağlı (kutup!); iki pil eşit aydınlanırsa LED'ler yanmaz, fan hızı şiddetle artar. Bir pil gölgelenince seride en zayıf hücre akımı sınırlar; fazla akım bypass-LED'den geçer → LED yanar, fan düşük gerilimle döner.

Evaluation item VII – annual energy yield and cost per kWh

Item VII (§6): Determine the annual energy yield and the cost of a kWh generated with a service life of 20 years and acquisition costs of 4000 EUR/kWp (including all expected maintenance costs) of the measured solar cell: a) by means of a radiation map; b) using a radiation map and Duisburg radiation distribution; c) using the specified parameters from Tarifa (Spain).

Formulas (manual §2.4, eq. 15–17)

$$E_{Jahr} = \int P_{Sonne}(t) \cdot \eta(t)\, dt \approx P_{Jahr} \cdot \eta$$ (a: constant η at 1000 W/m², eq. 16)

$$E_{Jahr} = \Sigma_i P_{Jahr} \cdot a_i \cdot \eta(S_i)$$ (b, c: η from your intensity series for every irradiance class S_i, eq. 17; a_i = share of the cumulative energy in the class)

P_{Jahr} (kWh/m² per year) from the radiation map of the appendix: Duisburg lies in the green/yellow-green area of the winter map (≈ 230–240 kWh/m² in October–March) and in the red/orange area of the summer map (≈ 760–770 kWh/m² in April–September) – read the exact colour yourself: **P_{Jahr}(Duisburg) ≈ 1000 kWh/m²/a**. Tarifa: 654.55 + 1264.67 = **1919.22 kWh/m²/a** (given).

Irradiance class S_X	Duisburg winter %	Duisburg summer %	Tarifa winter %	Tarifa summer %
up to 50 W/m²	28.5	16.5	11.4	8.9
up to 100 W/m²	27.4	13.2	9.2	6.6
up to 200 W/m²	24.0	18.6	14.7	9.8
up to 300 W/m²	11.5	13.1	12.3	8.7
up to 500 W/m²	7.2	18.8	22.0	15.0
up to 1000 W/m²	1.4	19.8	30.4	51.0

Use for every class a representative irradiance (e.g. the class mean 25, 75, 150, 250, 400, 750 W/m²) and take η(S) from your intensity series (interpolated; note that your simulator only reaches 300 W/m², so extrapolate carefully for 500–1000 W/m² and say so).

Worked example (η = 15 %, constant – variant a)

Quantity	Value
area for 1 kWp (STC: 1000 W/m²)	A = 1 kW / (1 kW/m² · 0.15) = 6.67 m²
Duisburg: yield per kWp	0.15 · 1000 kWh/m²/a · 6.67 m² = 1000 kWh/kWp/a
cost per kWh (20 years, 4000 €/kWp)	4000 € / (20 · 1000 kWh) = 0.20 €/kWh
Tarifa: yield per kWp	0.15 · 1919 · 6.67 = 1919 kWh/kWp/a
cost per kWh (20 years)	4000 € / (20 · 1919 kWh) = 0.10 €/kWh

Variant b/c are slightly lower, because η(S) is smaller at low irradiance (the weak-light shares dominate in Duisburg: 56 % of the winter energy comes in classes up to 100 W/m²). **Conclusion:** the same cell produces ≈ twice the energy in Tarifa and the kWh is about half as expensive – the location (irradiance and its distribution) matters as much as the efficiency. Degradation, inverter and maintenance are *not* included in the simple numbers (the manual says maintenance is part of the 4000 €/kWp).

TR E_Jahr = P_Jahr·η (sabit η) veya Σ P_Jahr·a_i·η(S_i). Duisburg ≈ 1000 kWh/m²/yıl, Tarifa 1919. 1 kWp = 6,67 m² (η = %15): kWh maliyeti 4000 €/(20·verim) → Duisburg ≈ 0,20 €, Tarifa ≈ 0,10 €.

LTspice – optional: solar-cell model (intensity and series resistance)

The manual does not require a simulation. The one-diode model (photocurrent source I_{ph} = C·35 mA, diode I_S = $1.4 \cdot 10^{-10}$ A with n = 1.2, R_S = 0.5 Ω, R_P = 500 Ω) reproduces Fig. C–E and shows cause and effect: Solarzelle_IU_Intensitaet.asc (C = 0.25 … 1) and Solarzelle_IU_Serienwiderstand.asc (R_S = 0.1, 2, 5 Ω) – or *04_LTspice/LTspice_starten.bat*. The .meas lines in the log give I_{SC} and P_{max}: C = 0.25/0.5/0.75/1 → I_{SC} = 8.7/17.5/26.2/35.0 mA, P_{max} = 3.4/7.5/11.6/15.8 mW; R_S = 0.1/2/5 Ω → P_{max} = 16.2/14.3/11.5 mW.

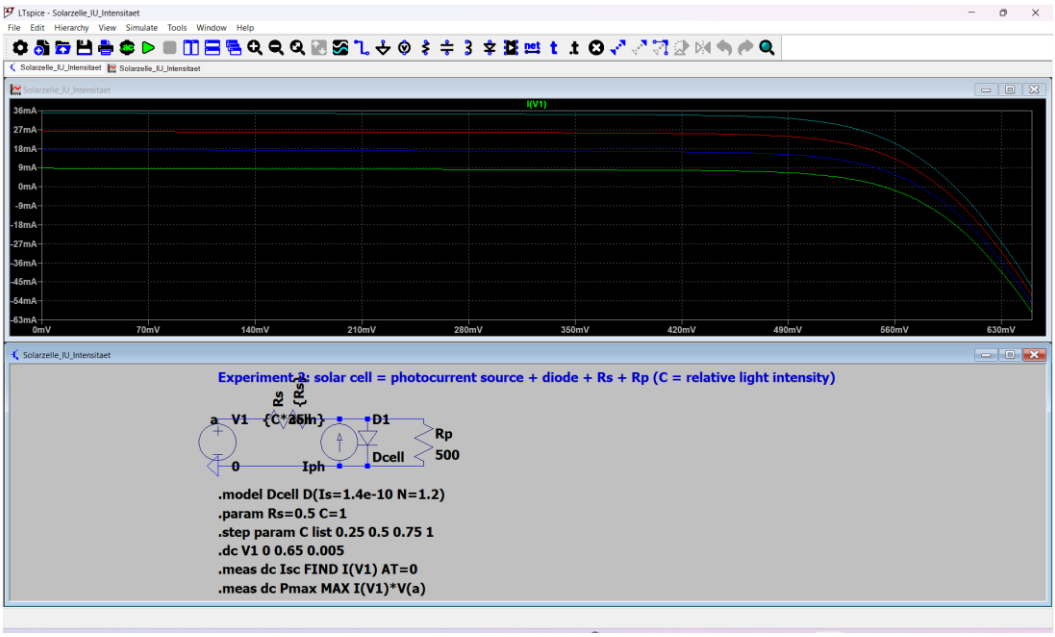


Fig. F – LTspice: four intensities (.step param C list 0.25 0.5 0.75 1).

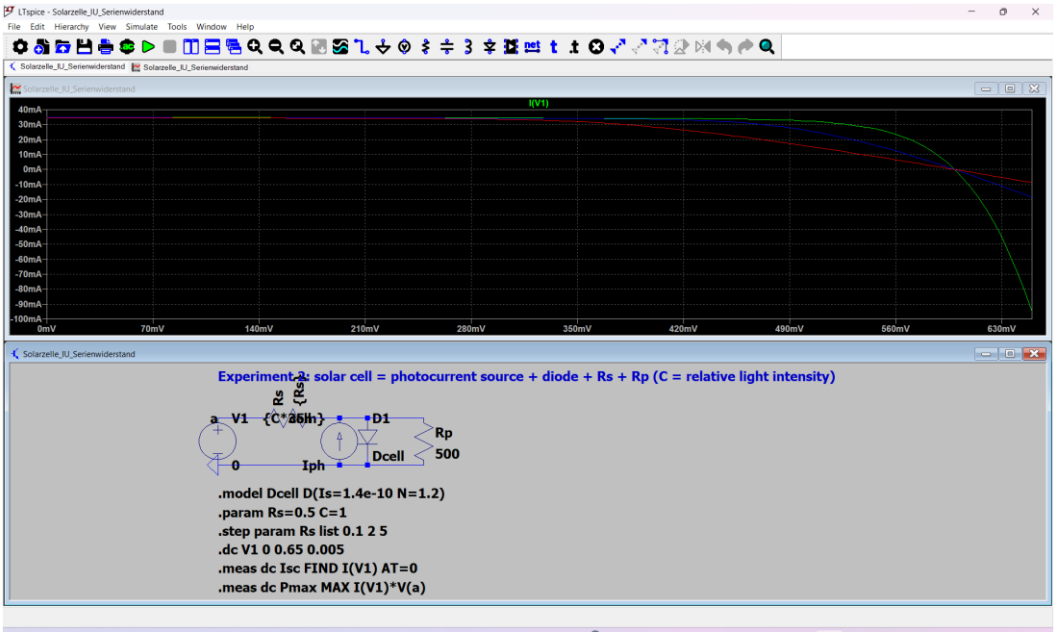


Fig. G – LTspice: series resistance steps (0.1, 2, 5 Ω): the curve rounds off and FF falls.

Antestat – likely questions with short answers

Question	Short answer
How does a solar cell work?	Photons with $E \geq W_G$ create e-h pairs; the pn field separates them → photocurrent; the load turns it into power.
Equivalent circuit?	Current source I_{ph} ‖ diode ‖ R_P, in series with R_S.
Define I_{SC}, U_{OC}, MPP, FF, η.	I at U = 0; U at I = 0; maximum of U·I; FF = $P_{MPP}/(U_{OC}I_{SC})$; η = P_{MPP}/P_{opt}.
How do I_{SC} and U_{OC} depend on intensity?	$I_{SC} \propto$ intensity; $U_{OC} \propto nU_T \ln(I_{SC}/I_S)$ – logarithmic.
How does temperature change a solar cell?	I_S ↑ exponentially → U_{OC} ↓ (≈ −2 mV/K), I_{SC} slightly ↑ (smaller W_G), η ↓.
What does R_S do? R_P?	R_S ↑: flatter curve near U_{OC}, FF ↓; R_P ↓: slope at U = 0, U_{OC} ↓ at low light, FF ↓.
How do you get the band gap from the dark curves?	$I_S(T) \propto \exp(-W_G/(nkT))$: plot ln I_S against 1/T, slope = $-W_G/(nk)$.
Why does a bad cell have a low fill factor?	Large R_S, small R_P, recombination (n ≈ 2).
Which loss mechanisms limit η?	Reflection, non-absorption ($E < W_G$), thermalisation, recombination, ohmic losses.
Why is there an optimum band gap?	Small W_G: many photons but small U_{OC} and thermalisation loss; large W_G: high U_{OC} but few absorbed photons; optimum ≈ 1.1–1.5 eV.
What is AM1.5G / STC?	Reference solar spectrum, 1000 W/m², cell temperature 25 °C; basis of the W_p rating.
What is a bypass diode for?	It carries the current of the string around a shaded cell (protects it from reverse stress, hot spots).
Why is the spectrum of the LED simulator a problem?	No infrared, blue peak: different from AM1.5G → spectral mismatch for Si.
How do you estimate the annual yield?	$E_{Jahr} = P_{Jahr} \cdot \eta$ (or the irradiance-class sum with η(S)); cost = 4000 €/kWp ÷ (20 years × kWh/kWp/a).

Lab-day checklist

- Antestat **14:00**, building **LT, room LT 227**, group 9, **3 February 2027**.
- **Plan with the supervisor** the temperatures (dark) and intensities (bright) so that W_G and η(C) can be evaluated (manual note under §5).

- Task order: (a) spectrum → (b) area → (c) dark curves at several T → (d) bright curves at 25/50/75/100 % → (e) fan circuit (observations table).
- Write down for every curve: temperature, intensity setting, sweep range, compliance current, area; save the data (USB stick) as tables U, I.
- Use the Excel template *06_Auswertung/Auswertung_Versuch2.xlsx* afterwards (enter data → FF, η, n, I_S, W_G, yield are computed).
- For the protocol: theory vs. measurement and error analysis for each item I–VII.